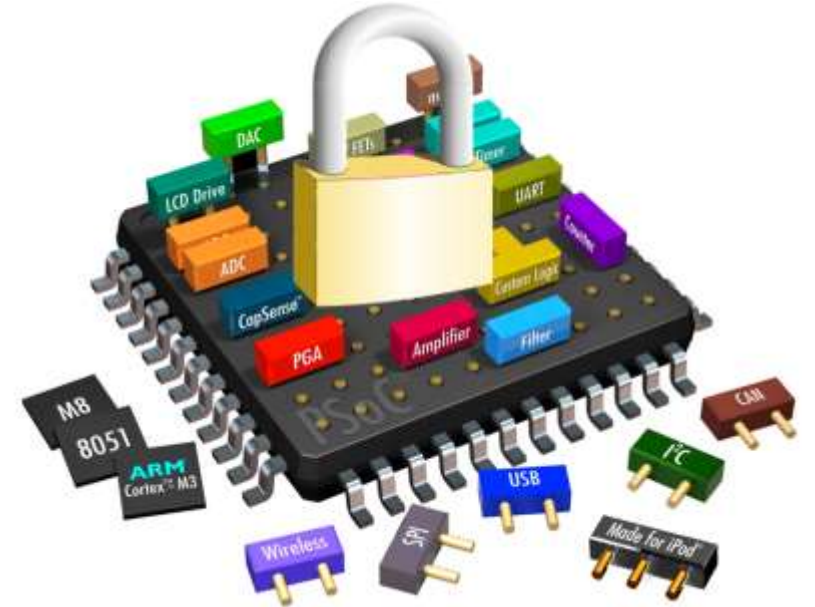


Hardware Security

TEE Examples: IBM PEF
(Protected Execution Facility)



Three Key Challenges in TEE Adoption

Balancing Isolation & Confidentiality

Server-class processors often link isolation and confidentiality, making TEEs expensive to operate.

Application Changes

Current implementations introduce many new security components requiring large application changes or greater trust.

Life Cycle Management

Managing secure entities in cloud deployments remains cumbersome due to attestation, secret provisioning, and integration complexity.

Protected Execution Facility (PEF)

- PEF provides a virtual machine-based TEE for OpenPOWER systems. It introduces the Protected Execution **Ultravisor**, new firmware managing hardware-enforced access control in a more privileged state than the hypervisor.
- PEF is supported in the latest POWER9 chip and demonstrates that combining access control for isolation with cryptography for confidentiality creates an effective confidential computing approach.

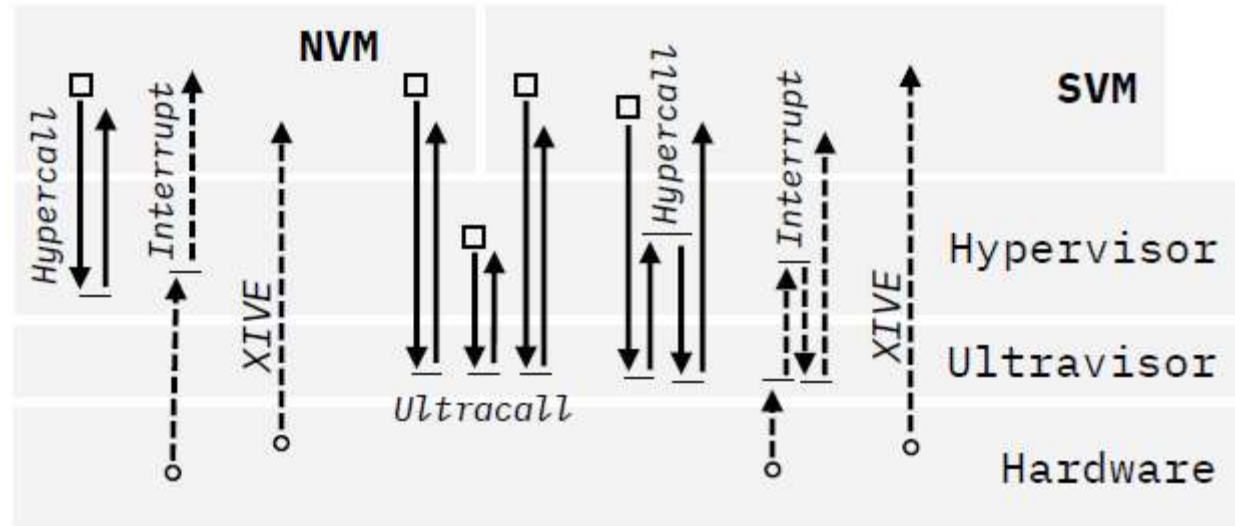
Secure State

Machine State Register (MSR)

Secure				Normal			
S	HV	PR		S	HV	PR	
1	0	0	privileged(OS)	0	0	0	privileged(OS)
1	0	1	problem	0	0	1	problem
1	1	0	ultravisor	0	1	0	hypervisor
1	1	1	(reserved)	0	1	1	problem (HV)

- PEF adds a new CPU state called secure state, more privileged than hypervisor state.
- The Ultravisor firmware manages all security-related hardware functions exclusively in this state.
- The Secure bit (S) in the Machine State Register determines whether system runs in normal or secure state.
- Secure state enables Protected Execution Ultravisor and Secure VMs.

Ultravisor



- Maintaining the **isolation** and **security** of the computation and associated data is the sole objective of the Ultravisor.
- System management continues to be the responsibility of the hypervisor.
- The hypervisor uses ultracalls to continue managing security sensitive facilities.
- In secure state, all hypervisor calls are delivered to the Ultravisor.
- Asynchronous interrupts are delivered to either the SVM or to the Ultravisor.

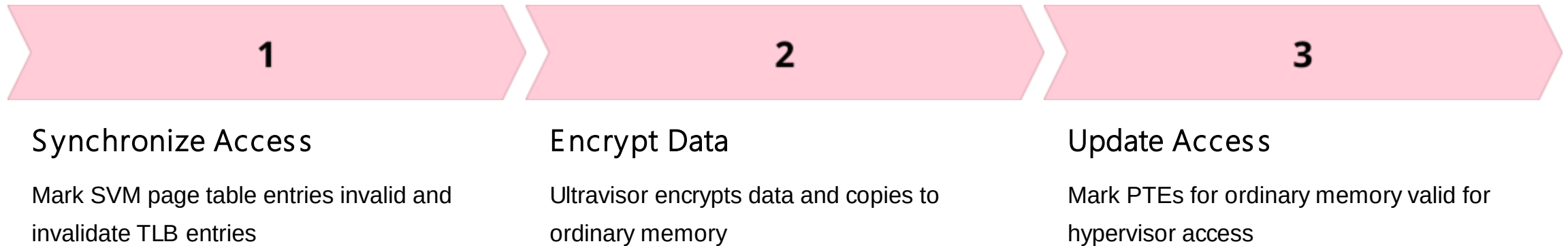
Access Control Architecture

- PEF uses security domains to enforce access control.
 - The hypervisor and normal VMs share one domain in normal memory.
 - Each Secure VM gets its own domain in secure memory.
- This provides bidirectional protection:
 - SVMs are protected from hypervisor and each other, while the hypervisor is protected from all SVMs.

Secure Memory Implementation

- Secure memory identified by MEMSM bit. In POWER9, implemented using high-order address bit RA(15).
- Hardware enforces that only processes in secure state can access secure memory.
- The Ultravisor controls the Partition Table Control Register, pointing to partition tables in secure memory.
- This isolates SVMs from each other, NVMs, and the hypervisor.

Maintaining Confidentiality



When hypervisor pages or dumps SVM, Ultravisor protects confidentiality through encryption with integrity using GCM before making data available.